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### Hairpin connector for traffic cable barrier system

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#### Abstract

A cable barrier assembly or system can include line posts extending along a barrier section. Cables can extend along the line posts. In addition, each line post can have a total number of cable retention features that is less than a total number of the cables. Each cable retention feature can include at least a partial loop that limits a range of radial motion of a cable extending through an eye of the partial loop, relative to an axis of the cable.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a division of and claims priority to and the benefit of U.S. application Ser. No. 17/006,207, filed Aug. 28, 2020, which is incorporated herein by reference in its entirety.

### BACKGROUND

(1) The present disclosure relates to barriers and safety systems and, more particularly, to cable safety systems. Cable barrier systems are often employed to redirect motor vehicles toward a less hazardous path. Often, cable barrier systems are utilized along roadways and in the medians between roadways. For example, cable barrier systems can be utilized to redirect an errant motor vehicle headed through a median toward oncoming traffic back into the intended direction of travel or prevent a roadway departure on the side of a road.

(2) Cable barrier systems employ high tension, steel cables supported and restrained on posts mounted in the ground. During a vehicle impact with the safety system, the posts and cables work together to redirect the vehicle. It is important that the system is sturdy enough redirect the vehicle, yet not so rigid that it causes extraordinary harm to the occupants of the vehicle. Often during a collision event, the posts impacted are bent and the cables supported and restrained by the impacted post can be pulled downward with the impacted post, which can reduce the effectiveness of the cable barrier system. Some cable barrier systems employ release mechanisms to release the cables from post restraint upon impact. Although such systems are workable, improvements in cable barrier systems continue to be of interest.

### SUMMARY

- (3) Embodiments of a cable barrier assembly or system can include a plurality of line posts extending along a barrier section. A plurality of cables can extend along the line posts. In addition, each line post can have a total number of cable retention features that is less than a total number of the cables. Each cable retention feature can include at least a partial loop that limits a range of radial motion of a cable extending through an eye of the partial loop, relative to an axis of the cable.
- (4) Other embodiments of a cable barrier assembly can include a plurality of line posts extending along a barrier section. A plurality of cables can extend along the line posts comprising first, second, third and fourth cables. The first and second cables of the plurality of cables can be aligned in vertical elevation and touching between adjacent ones of the line posts. In addition, the first and second cables can be not aligned in vertical elevation and not touching at the line posts, such that the first cable is separated from the second cable at the line posts.
- (5) Another version of a cable barrier assembly can include a plurality of line posts extending along a barrier section. A plurality of cables can extend along the line posts. At least some of the cables can be secured to each line post. However, not every line post is secured to all of the cables, in some versions.
- (6) Embodiments of a method for releasably connecting cables to posts in a roadway cable barrier system also is disclosed. The method can include mounting a first cable to a first post such that the first cable is substantially restrained in all radial directions other than upper vertical directions. The method also can include extending a second cable adjacent to the first post such that the second cable is unrestrained in upper vertical radial directions and in other radial directions. In addition, the method can include mounting a second cable to a second post such that the second cable is substantially restrained by the second post. Versions of the method can include extending the first cable adjacent to the second post such that the first cable is substantially unrestrained by the second post.
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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

- (1) For a detailed description of example embodiments, reference will now be made to the accompanying drawings in which:
- (2) FIG. 1 is a schematic sectional side view of an embodiment of a cable barrier system.
- (3) FIG. 2 is a schematic top view of a portion of an embodiment of a cable barrier system.
- (4) FIG. 3 is a schematic sectional side view of another embodiment of a cable barrier system.
- (5) FIG. 4A is an isometric view of a portion of an embodiment of the cable barrier system of FIG. 3.
- (6) FIGS. 4B-4E are enlarged, sectional axial views of embodiments of the cable barrier system of FIG. 4A at different areas thereof.
- (7) FIG. 4F is an exaggerated, schematic side view of a portion of the cable barrier system of FIG. 4A.
- (8) FIG. 5 is an isometric view of an embodiment of a hairpin connector.
- (9) FIG. 6 is a side view of the connector of FIG. 5.
- (10) FIG. 7 is a front view of the connector of FIG. 5.
- (11) FIG. 8 is a sectional axial view of an embodiment of a cable barrier system through a line post.

### **DEFINITIONS**

- (12) Various terms are used to refer to particular system components. Different companies may refer to a component by different names—this document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be

interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections.

#### DETAILED DESCRIPTION

(13) The following discussion is directed to various embodiments of the invention. Although one or more of these embodiments may be preferred, the embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to intimate that the scope of the disclosure, including the claims, is limited to that embodiment.

(14) FIG. 1 is a schematic sectional view of a cable barrier assembly or system, identified by the numeral **10**. Cable barrier system **10** can include cables **12a**, **12b**, and **12c** (collectively, cables **12**) held in tension from a terminal end **14** through a barrier section, or length of need **16**. Cable barrier system **10** may include additional terminal ends **14** and intermediate terminal ends (not shown). System **10** is illustrated and described herein as an example of a three-cable, highway median safety barrier, or cable guardrail. However, it should be realized that the various systems, assemblies, members and concepts described herein may be utilized in various installations and configurations for varying purposes. It should further be understood that various components of these embodiments can be utilized with various types and designs of barrier systems including, but not limited to, cable barrier systems, W-beam guardrail systems, crash cushions and attenuators.

(15) Embodiments of terminal end **14** can include a cable-release anchor **17** having a leveraging member **18** and foundation **36**, one or more weak terminal posts **20**, and one or more standard terminal posts **22**. The terminal ends **26** of cables **12** can be removably mounted to cable-release anchor **17** substantially at ground level **28** and removably connected to terminal posts **20**, **22** and line posts **24** of length of need (LON) section **16**. The term ‘substantially’ can be defined as to a great or significant extent, for the most part, or essentially. Cables **12** can have turnbuckles **30** and can be angled upward relative to ground level **28** through a portion of terminal end section **14** until the desired distance above ground level **28** is obtained. Terminal end **14** can be a gated terminal wherein substantially no resistance is provided upon impact by an errant vehicle.

(16) In some versions, the length of need section **16** can include a plurality of spaced line posts **24**. Cables **12** can be removably connected to line posts **24** in tension. The length of need **16** may be any desired length. The system **10** can include a variety of fittings, such as the cable splice fittings disclosed in U.S. Pat. No. 9,228,306, issued on Jan. 5, 2016, which is incorporated herein by reference in its entirety. Any of the systems, methods and components disclosed in U.S. Pat. No. 9,228,306 can be adapted and employed in this disclosure.

(17) Embodiments of the cables **12** can be releasably connected to terminal posts **20**, **22** and line posts **24** in a manner such that when an individual post fails and is moved toward the ground level **28**, the cables **12** are released from that individual post. For example, if a vehicle **38** (FIG. 2) impacts cable barrier system **10** in length of need section **16** and collapses one line post **24** toward the ground level **28**, cables **12** can be released from that line post **24** so that cables **12** remain supported above ground level **28** and in contact with the vehicle **38** and do not go under the vehicle. The cables **12** can remain supported above ground level by the remaining portion of the cable barrier system **10**, thereby urging the vehicle **38** back to its designated and desired path (i.e., roadway).

(18) In FIG. 2, the system **10** illustrates one example of mounting the system **10** for absorbing impact from the errant vehicles **38** and redirecting them from two directions, such as for highway medians. Arrows **38** illustrate the direction of travel of vehicles impacting system **10**. Posts **20**, **22**, and **24** can each have a face **20a**, **22a**, and **24a** respectively, adapted for removably mounting

cables **12**. The post faces **20a**, **22a**, and **24a** can be desirably oriented to face oncoming vehicles such that cables **12** are positioned between posts **20**, **22**, and **24** and the direction of vehicle travel **38**. For applications wherein it is probable that vehicles may impact from either direction, posts **20**, **22**, and **24** may be installed such that at least a portion of post faces **20a**, **22a**, and **24a** are oriented toward oncoming traffic. In the illustrated embodiment, posts **20**, **22**, and **24** are installed with each post face oriented opposite the orientation of the adjacent post faces, although this is not necessarily required.

(19) FIGS. **3** and **4A-4F** depict embodiments of a four-cable barrier assembly or system **100**. Examples of the cable barrier system **100** can include one or more of an anchor **117**, a weak terminal post **120**, and a standard terminal post **122**. The terminal ends of cables **112** can be removably mounted to the anchor **117** and removably connected to terminal posts **120**, **122** and line posts **124**. The cables **112** can be angled upward relative to ground level **128** until the desired distance above ground level **128** is obtained. As shown in FIG. **4A**, the cables **112** can alternate on opposite lateral sides of adjacent ones of the line posts **124** (e.g., line posts **124a**, **124b**, **124c**, **124d**). In another version, the cables **112** can all be the same sides of the line posts **124**.

(20) Embodiments of each line posts **124** can include a post body **130** (FIG. **4A**) secured in the ground as is known in the art. The post body **130** can include an elongated, rectangular profile with an open top **132**. Examples of the post body **130** can have a hollow interior and a slot **134** extending along one side of the post body **130**. In one version, the slot **134** can extend for the entire length of the post body **130**.

(21) In some versions, a removable connector or hairpin **140** (FIGS. **5-7**) can be installed through the top **132** of the line post **124** and into the interior of the post body **130**. The hairpin **140** can include cable retention features, such as partial loops or loops **142** that extend to the exterior of the post body **130** through the slot **134**. The loops **142** can be used to capture and restrain the cables **112** as described herein. The illustrated example of hairpin **140** can include a lower loop **142a**, a middle loop **142b** and a top loop **142c**. The hairpin **140** also can include a hook **144** at its upper end for engaging and resting on the top **132** of the post body **130**. Versions of the hairpins **140** can be constructed from steel or other materials, such as aluminum, to reduce their weight and airborne environmental impact during a vehicle collision with the cable barrier system **100**.

(22) In the versions of FIGS. **4A-4F**, the cable barrier system **100** can include four cables **112**, including a lower cable **112a**, a middle cable **112b** and two top cables **112c** and **112d**. In one example, at least two of the cables **112a**, **112b** are parallel (or substantially parallel, meaning not perfectly parallel in a mathematical sense) to each other along the barrier section. In another example, at least two other ones of the cables **112c**, **112d** are not parallel to each other (e.g., they can be skewed) along the barrier section. As shown in FIG. **8**, embodiments of the cable barrier assembly **100** can include at least some of the cables **112** being secured to each line post **124**, and not every line post **124** securing all of the cables **112**.

(23) In another example, cables **112c**, **112d** can be horizontally aligned (FIGS. **4B** and **4D**) in vertical elevation between adjacent ones of the line posts **124**. Some examples of the cables **112c**, **112d** are not horizontally aligned (FIGS. **4C** and **4E**) in vertical elevation, such as at the line posts **124**. Accordingly, the cables **112c**, **112d** can be vertically separated from each other (FIGS. **4C** and **4E**) at the line posts **124**. In one embodiment, at the line posts **124**, the cables **112c**, **112d** can be vertically aligned with each other (FIGS. **4C** and **4E**), such that they are not horizontally offset from each other. Alternatively, at the line posts **124**, the cables **112c**, **112d** can be not vertically aligned with each other, such that they are horizontally offset from each other.

(24) Embodiments of the cable barrier assembly **100** can include the cables **112c**, **112d** alternating at two different vertical elevations at adjacent ones of the line posts **124**. In one version, the cables **112c**, **112d** can be characterized as not being intentionally horizontal (FIG. **4F**) along the barrier section (or intentionally perpendicular to line posts **124**), except where they incur inflections at the line posts **124**. In contrast, embodiments of the cables **112a**, **112b** can be intentionally

perpendicular to posts **124**, and horizontal and/or parallel to the ground **128**.

(25) Another version of the cable barrier assembly **100** can include each line post **124** with a total number of cable retention features (e.g., three loops **142**) that is less than a total number of the cables **112** (e.g., four cables **112**). Examples of each cable retention feature can include at least a partial loop or a loop **142** that limits a range of radial motion of a cable **112** extending through an eye of the loop **142**, relative to an axis of the cable **112**.

(26) Embodiments of the cable barrier assembly **100** can include two top cables **112c**, **112d** at a same or substantially same vertical elevation, such as in a plane (e.g., a horizontal or substantially horizontal plane). The two top cables **112c**, **112d** can be side-by-side, or substantially side-by-side, when away from the line posts **124**. Other versions can include a center-to-center spacing **114** (FIGS. **4B** and **4C**) between the two top cables **112c**, **112d**, when assembled or installed. For example, the spacing **114** can be not greater than one outer diameter (e.g., 0.75 inches) of the cables **112**. In another example, the spacing can be not greater than 1.5 outer diameters, not greater than 2 outer diameters, or even not greater than 4 outer diameters of the cables **112**.

(27) In one example (e.g., FIG. **4B**), at least portions of the two top cables **112c**, **112d** can extend parallel or semi-parallel to each other. The term ‘semi-parallel’ can mean substantially, essentially or mostly parallel. In some versions, the two top cables **112c**, **112d** can be touching at the same or substantially same vertical elevation (e.g., FIG. **4B**), or touching at different vertical elevations (e.g., FIG. **4D**). In some versions, the two top cables **112c**, **112d** can be separated and in vertical alignment (e.g., FIG. **C**), or separated and not vertically aligned (e.g., FIG. **4E**). Examples of the two top cables **112c**, **112d** can be touching between adjacent ones of the line posts **124**, and not touching at the line posts **124**, such that the two top cables **112c**, **112d** are separated, such as vertically separated, from each other at the line posts **124**. Between the line posts **124**, embodiments of the cables **112a**, **112b** can be not parallel to the cables **112c**, **112d**. At the line posts **124**, at least portions of the cables **112c**, **112d** can be parallel to the cables **112a**, **112b**.

(28) In another example, the two top cables **112c**, **112d** do not twist with each other. The term ‘twist’ can be defined as helical or semi-helical interplay. For example, ‘one complete twist’ can include 360° of helical interplay between or winding of two cables along a helical axis. In some versions (compare FIGS. **4B-4E**), the two top cables **112c**, **112d** twist for not greater than about 90° of twist, not greater than about 135° of twist, or even not greater than about 180° of twist.

(29) In some versions, the two top cables **112c**, **112d** can alternate (compare FIGS. **4C** and **4E**) extending (a) through the cable retention features (e.g., loops **142c**) on adjacent ones of the line posts **124**, and (b) externally and adjacent to the cable retention features (e.g., loops **142c**) on adjacent ones of the line posts **124**. Examples of the two top cables **112c**, **112d** can alternate being vertically above and below each other at adjacent ones of the line posts **124**.

(30) Accordingly, at each line post **124**, one of the cables **112c**, **112d** can comprise a ‘free cable’ that is substantially unrestrained (i.e., not captured by a cable retention feature). For example, in FIG. **4C** cable **112c** is the free cable, whereas cable **112d** is the free cable in FIG. **4E**. Even though the free cable is substantially unrestrained, it can be located adjacent to an exterior of a cable retention feature. Moreover, the free cable can have a range of freedom of radial motion. For example, the range of freedom of radial motion can be about 90° to about 135° (see FIG. **8**), relative to an axis of the free cable.

(31) Other embodiments can include a method for releasably connecting cables to posts in a roadway cable barrier system. For example, the method can include mounting a first cable to a first post such that the first cable is substantially restrained in all radial directions other than upper vertical directions (e.g., not axially restrained or upwardly restrained). That is to say, the eye of loop **142** through which the first cable extends restrains the cable from movement in radial directions extending outward from the long central axis of the first cable; however, because the hairpin **140** in the example embodiment is suspended within the line post **124**, movement of the cable vertically may still be possible in spite of the eye of the loop **142** because the entire hairpin

**140** may move vertically (and relative to line post **124**) if sufficient upward force is applied.

Versions of the method can include extending a second cable adjacent to the first post such that the second cable is unrestrained in upper vertical radial directions and in other radial directions. The second cable could be not touching or free of contact with the line post or cable retention feature. The method also can include mounting the second cable to a second post such that the second cable is substantially restrained by the second post. In addition, the method can include extending the first cable adjacent to the second post such that the first cable is substantially unrestrained by the second post. The method can be repeated for subsequent posts after the second post.

(32) Referring again to FIGS. 5-7, embodiments of the hairpin **140** can have a unique geometry that is flatter and has less deviation at its bottom than existing hairpin designs. For example, the hairpin **140** can include an elongated body **146** extending in a single plane and comprising a top (e.g., hook **144**), and a plurality of cable retention features (e.g., loops **142**), and a lower body **148** that is not co-planar with the elongated body **146**. The lower body **148** can comprise a flare extending in only one direction. Versions of the flare can extend at a single, out-of-plane angle. Examples of the single, out-of-plane angle can be about 6° with a tolerance of +5° and -0°. Embodiments of the hairpin **140** can comprise an approximately 7/16" round wire, and examples of the flare can have an out-of-plane dimension of about 9/16" or more. In some versions, the hairpin **140** can have an overall length, and the flare can comprise not greater than about 10% of the overall length, such as not greater than about 5%, and at least about 1% of the overall length of the hairpin **140**.

(33) Many of the connections in the drawings are shown as direct couplings having no intervening devices, but not expressly stated as such in the description above. Nevertheless, this paragraph shall serve as antecedent basis in the claims for referencing any connection as "directly coupled" for connections shown in the drawing with no intervening device(s).

(34) The above discussion is meant to be illustrative of the principles and various embodiments of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

## Claims

1. A hairpin connector and post for a traffic cable barrier system, comprising: an elongated body extending in a single plane and comprising a top at a proximal end, the top is configured to rest on a top of the post, the elongated body is configured to slidably and releasably engage an interior of the post, at least three cable retention features extend from the elongated body including a top loop, a middle loop and a lower loop that are configured to extend from the interior of the post to an exterior of the post to capture and restrain respective cables of the traffic cable barrier system, and a lower body at a distal end of the elongated body; and a flare at an end of the lower body, the flare extending in only one direction, and wherein the flare extends laterally outward relative to a longitudinal plane of the elongated body.
2. The hairpin connector and post of claim 1, wherein the flare extends at a single, out-of-plane angle relative to the elongated body.
3. The hairpin connector and post of claim 2, wherein the single, out-of-plane angle is in a range of 6° to 11°.
4. The hairpin connector and post of claim 1, wherein the hairpin connector comprises a 7/16" round wire, and an out-of-plane dimension between an outer surface of the elongated body and an end of an outer surface of the flare is 9/16".
5. The hairpin connector and post of claim 1, wherein the hairpin connector has an overall length, and the flare has a flare length that is not greater than 10% and at least 1% of the overall length of the hairpin connector.



6. A hairpin connector and post for a traffic cable barrier system, comprising: an elongated body extending in a single plane and comprising a top at a proximal end, the top is configured to rest on a top of the post, the elongated body is configured to slidably and releasably engage an interior of the post, cable retention features extending from the elongated body and configured to extend from the interior of the post to an exterior of the post to capture and restrain respective cables of the traffic cable barrier system, the cable retention features are spaced apart from each other by straight portions of the elongated body, and a lower body including another straight portion at a distal end of the elongated body opposite the top; and a flare extending from the lower body, wherein the flare extends laterally outward relative to a longitudinal plane of the elongated body.
  7. The hairpin connector and post of claim 6, wherein the cable retention features comprise a top loop, a middle loop and a lower loop.
  8. The hairpin connector and post of claim 6, wherein the flare extends in only one direction.
  9. The hairpin connector and post of claim 6, wherein the flare extends at a single, out-of-plane angle relative to the elongated body.
  10. The hairpin connector and post of claim 9, wherein the single, out-of-plane angle is in a range of 6° to 11°.
  11. The hairpin connector and post of claim 6, wherein the hairpin connector comprises a 7/16" round wire.
  12. The hairpin connector and post of claim 6, wherein an out-of-plane dimension between an outer surface of the elongated body and an end of an outer surface of the flare is 9/16".
  13. The hairpin connector and post of claim 6, wherein the hairpin connector has an overall length, and the flare has a flare length that is not greater than 10% and at least 1% of the overall length of the hairpin connector.
  14. The hairpin connector and post of claim 6, wherein all of the straight portions are co-linear.
  15. The hairpin connector and post of claim 6, wherein there is no straight portion between the top and a first one of the cable retention features.
  16. A hairpin connector for a traffic cable barrier system, the hairpin comprising: an elongated body extending in a single plane and comprising a top and cable retention features extending from the elongated body, the cable retention features are spaced apart from each other by straight portions of the elongated body, a lower body including another straight portion opposite the top, and the top comprises a top straight portion that extends in a same direction as the straight portions, but the top straight portion is not co-linear with the straight portions; and a flare extending from the lower body, wherein the flare extends laterally outward relative to a longitudinal plane of the elongated body.
  17. The hairpin connector of claim 16, wherein the cable retention features comprise a top loop, a middle loop and a lower loop, and the flare extends in only one direction.
  18. The hairpin connector of claim 16, wherein the flare extends at a single, out-of-plane angle relative to the elongated body.
  19. The hairpin connector of claim 16, wherein the hairpin connector has an overall length, and the flare has a flare length that is not greater than 10% and at least 1% of the overall length of the hairpin connector.
  20. The hairpin connector of claim 16, wherein there is no straight portion between the top and a first one of the cable retention features.
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